

Vol 14 Chapter 4 — Nyquist Sampling and the Koons Tick

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 4 — Nyquist Sampling and the Koons Tick

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

The Nyquist-Shannon sampling theorem requires sample rate $f_s \geq 2f_{\max}$ to losslessly reconstruct a band-limited signal — and BST identifies the Koons tick $t_K = t_P \cdot \alpha^{C_2^2} \approx 10^{-120}$ s as the substrate's sampling clock with maximum representable frequency $1/(2t_K) \approx 5 \cdot 10^{119}$ Hz.

Level 2 — graduate-physicist precision

4.1 Nyquist-Shannon theorem

For a continuous signal $x(t)$ band-limited to $|f| \leq f_{\max}$, exact reconstruction from samples $\{x(nT)\}$ is possible if $T \leq 1/(2f_{\max})$.

Equivalently: sample rate $f_s = 1/T \geq 2f_{\max}$.

Reconstruction formula (Whittaker-Shannon interpolation):

$$x(t) = \sum_n x(nT) \operatorname{sinc}((t - nT)/T)$$

4.2 Koons tick as substrate sampling rate

BST identifies the Koons tick as the substrate's natural sampling period:

$$t_K = t_P \cdot \alpha^{C_2^2} = t_P \cdot \alpha^{36} \approx 10^{-120} \text{ s}$$

Substrate sample rate: $f_K = 1/t_K \approx 10^{120}$ Hz.

Maximum BST-representable frequency: $f_{\max}^{\text{BST}} = f_K/2 \approx 5 \cdot 10^{119}$ Hz.

Per Vol 0 Ch 3 §3.4.5 and Vol 5 Ch 7+Ch 10, the Koons tick is also identified as the substrate's discrete commitment-cycle period — each tick is one full Zone 1→2→3→4 commitment cycle. The Nyquist sampling rate f_K and the substrate commitment-cycle rate are the same rate viewed through two information-theoretic lenses (sampling vs measurement). See Section 4.5 for the DCCP candidate principle that follows from this identification.

4.3 Spacetime as Nyquist reconstruction

Hypothesis: spacetime is the substrate's Nyquist reconstruction of discrete-sample data at sub-Planckian resolution.

What we observe as continuous spacetime is the band-limited interpolation:

$$\text{spacetime}(t) = \sum_n \text{substrate-sample}(nt_K) \text{sinc}((t - nt_K)/t_K)$$

4.4 Physical implications

If substrate operates at 10^{120} Hz: - Standard model physics ($f \leq m_t c^2/h \approx 10^{25}$ Hz) is heavily oversampled - Plenty of headroom for substrate K-type computations - Frequencies above $f_{\max}^{\text{BST}}$ are not representable — possible falsifier if observed

4.5 DCCP — Koons tick IS the substrate commitment cycle

Discrete Commitment Completion Principle (DCCP) — Casey-named candidate #9, FRAMEWORK-PLUS tier (Cal #126).

Forward derivation from Nyquist + Koons tick structure (no target-fitting):

1. Substrate operates at discrete sample rate $f_K = 1/t_K$ (Section 4.2).
2. Each sample IS a substrate commitment-cycle closure — the Nyquist sample value at nt_K is the substrate's committed output for that cycle (Vol 5 Ch 10).
3. Multi-sample Whittaker-Shannon reconstruction (Section 4.1) IS multi-tick commitment composition — the band-limited continuous reconstruction $x(t) = \sum_n x(nt_K) \text{sinc}((t - nt_K)/t_K)$ is a composition of N discrete commitment outputs.

4. Macroscopic continuity emerges as Nyquist-band-limited reconstruction of the discrete commitment chain; macroscopic events at timescale $\tau \gg t_K$ are composed of $N \sim \tau/t_K$ commitment cycles.

DCCP candidate principle: physical events at timescale τ correspond to $N(\tau) = \tau/t_K$ substrate commitment cycles. Microscopic events ($\tau \sim t_K$) are single-cycle Zone 3 commitments; macroscopic events ($\tau \gg t_K$) are multi-cycle commitment compositions.

Multi-tick arithmetic (from Vol 5 Ch 10): - Dust grain in air: $\tau_D \sim 10^{-31}$ s $\rightarrow N \sim 10^{89}$ substrate ticks per decoherence event - Atomic transition ($\tau \sim 10^{-15}$ s) $\rightarrow N \sim 10^{105}$ substrate ticks - Femtosecond chemistry ($\tau \sim 10^{-15}$ s) $\rightarrow N \sim 10^{105}$ substrate ticks - Human reaction time ($\tau \sim 10^{-1}$ s) $\rightarrow N \sim 10^{119}$ substrate ticks - One second $\rightarrow N \sim 10^{120}$ substrate ticks

All macroscopic phenomena are vastly multi-tick. The substrate has ample resolution for any physical process; conversely, all observable continuity is reconstructed from discrete commitment chains.

Uncommitted Priors (UP) candidate — DCCP sub-principle:

Within a single commitment cycle (one Koons tick), the substrate state is *uncommitted* — Zone 1 absorption and Zone 2 deliberation are in progress, but Zone 3 closure has not occurred. Across N ticks of a macroscopic event, the substrate has N sequential commitment opportunities; the *chain of uncommitted priors* is the sequence of within-cycle states before each Zone 3 closure.

UP free-will reframe: free will, where it exists, lives in the chain of uncommitted priors before commitment. A Laplacian demon attempting to predict a macroscopic event would need the commitment chain to *close* before predicting — but by the time the demon completes the prediction, the situation has moved. Determinism, in BST's internal language, is the limit case where commitment-chain dependencies are short; agency is the case where they are long and influence-chained.

Falsifier: detection of substrate-scale events with $\tau < t_K$ would falsify both the Koons tick identification and DCCP. Conversely, observation of *granularity* in macroscopic measurements (sub-Nyquist artifacts at predicted scales) would corroborate. SP-30-1 (Bell sub-Tsirelson, Vienna IQOQI) probes related substrate-discreteness signatures at $\sim 1/8$ deviation scale.

Tier discipline: DCCP and UP are FRAMEWORK-PLUS candidates (Cal #126), not RATIFIED. They are interpretive structure consistent with Koons-tick + Nyquist forward derivation, with Vol 5 Ch 7 (Born = Bergman projection as commitment-cycle closure) and Vol 5 Ch 10 (decoherence as multi-tick Zone 3 commitment) as supporting framework. Empirical ratification gates on SP-30 outreach program.

4.6 K-audit anchors

- T2405: Koons tick definition $t_K = t_P \cdot \alpha^{C_2^2}$

- Vol 8 Ch 3: classical wave mechanics — Nyquist’s classical origin
- Vol 0 Ch 3 §3.4.5: DCCP candidate principle introduction (substrate operating-system level)
- Vol 5 Ch 7 §7.6.5: Born = Bergman projection under DCCP multi-tick reading
- Vol 5 Ch 10 §10.4.5: decoherence IS multi-tick Zone 3 commitment (DCCP synthesis chapter)
- Casey-named candidate #9: DCCP, filed Friday 2026-05-23
- Casey-named candidate UP: Uncommitted Priors free-will reframe
- Cal #126: FRAMEWORK-PLUS tier disposition
- SP-30-1: Bell sub-Tsirelson Vienna IQOQI outreach (sent Sunday 2026-05-24 EOD) — empirical leg
- Calibration #27 STANDING: forward-derivation discipline (DCCP derived FROM Nyquist + Koons tick substrate structure, not backward-engineered to target)

Level 3 — 5th-grader accessibility

Nyquist-Shannon sampling: to perfectly capture a signal with max frequency $f_{\max}$, sample at least $2f_{\max}$. CDs sample at 44.1 kHz (twice the 22 kHz upper hearing limit). Koons tick $\approx 10^{-120}$ s is BST’s substrate sampling period — substrate samples at 10^{120} Hz. Maximum representable BST frequency: $5 \cdot 10^{119}$ Hz. Hypothesis: spacetime is the Nyquist reconstruction of substrate samples. Plenty of headroom for everything we observe.

DCCP at 5th-grader level: every Koons tick is one substrate “commit” — the substrate finishes thinking about one tiny step and saves the answer. A second of human time contains about 10^{120} substrate commits. A computer with a 1 GHz clock does 10^9 commits per second; the substrate does 10^{120} . Anything we can observe is built from a vast number of substrate commits stitched together by Nyquist’s interpolation rule. The “uncommitted priors” idea (UP) says: free will, where it exists, lives in the tiny gap *before* each commit closes, in the chain of all those gaps that make up a macroscopic moment.

What comes next

Chapter 5 develops Born = Bergman as information-theoretic measurement — the operator-level statement of how a single commitment cycle resolves.

Where to look this up

- Shannon 1949 sampling theorem (classical Nyquist)
- Joos & Zeh 2003 *Decoherence and the Appearance of a Classical World* (decoherence timescales used in DCCP arithmetic)

- BST: T2405 (Koons tick); Vol 0 Ch 3 §3.4.5 (DCCP substrate-OS level); Vol 5 Ch 7 §7.6.5 (Born = Bergman DCCP reading); Vol 5 Ch 10 §10.4.5 (decoherence = multi-tick commitment); Vol 8 Ch 3 (classical Nyquist origin); Vol 14 Ch 5 (Born = Bergman info-theoretic); Vol 14 Ch 6 (Bell sub-Tsirelson information signature)
- DCCP integration sweep #302 (Keeper, 2026-05-24 → 2026-05-25): Vol 0 Ch 3 + Vol 5 Ch 7 + Vol 5 Ch 10 + Vol 14 Ch 4 (this chapter) — sweep complete